Project #4 guidelines for Python

If you decide to write your project using python, I'll tell you where to implement the code in project #4 framework.

Implementation

a. Use your custom data preprocessing methods

a. Complete ***Preprocessor class*** in <u>*preprocessor.py*</u>
   Use your custom data preprocessing functions in project #1 as the methods in Preprocessor class.

preprocessor.py

preprocessor.py > ...

```python
# preprocessor.py


class Preprocessor:
    """ TODO """
    def __init__(self, df):
        # Initialize the preprocessor with a DataFrame
        self.df = df

```

b. Complete ***DataPreprocessing function*** in <u>*main.py*</u>
   Use the custom methods in project #1 you implemented to complete dataPreprocessing function.

main.py

main.py > main

```python
import numpy as np
import pandas as pd
from preprocessor import Preprocessor
from model import DecisionTreeClassifier
import os


def dataPreprocessing():
    """ TODO, use your own dataPreprocess function here. """

    return


```

Note: You can either use your data preprocessing methods in project #1 or implement a new one for project #4.

b. Complete KNN in *model.py*

a. Implement necessary methods in *KNearestNeighborClassifier* (You don't have to follow the template to implement your KNN)

```python
# K-Nearest Neighbors Classifier
class KNearestNeighborClassifier(Classifier):
    def __init__(self, k=3, distance_metric='euclidean', p=3):
        #TODO
        pass

    def fit(self, X, y):
        #TODO
        pass

    def predict(self, X):
        y_pred = [self._predict(x) for x in X]
        return np.array(y_pred)

    def predict_proba(self, X):
        #TODO
        pass

    def _predict(self, x):
        #TODO
        pass
```

c. Complete main function in *main.py*

a. Build your KNN and save the predict label as .csv file

```python
def main():
    root_path = r"yourtraining_data" # change the root path
    train_X = os.path.join(root_path, "train_x.csv")
    train_y = os.path.join(root_path, "train_y.csv")
    test_X = os.path.join(root_path, "test_x.csv")

    model = KNearestNeighborClassifier()
    model.fit(train_X,train_y)

    # TODO
    # build your KNN model
    # predict the output of the testing data
    # remember to save the predict label as .csv file
```